DTC P0A8D:00 [SKYACTIV-D 1.5]

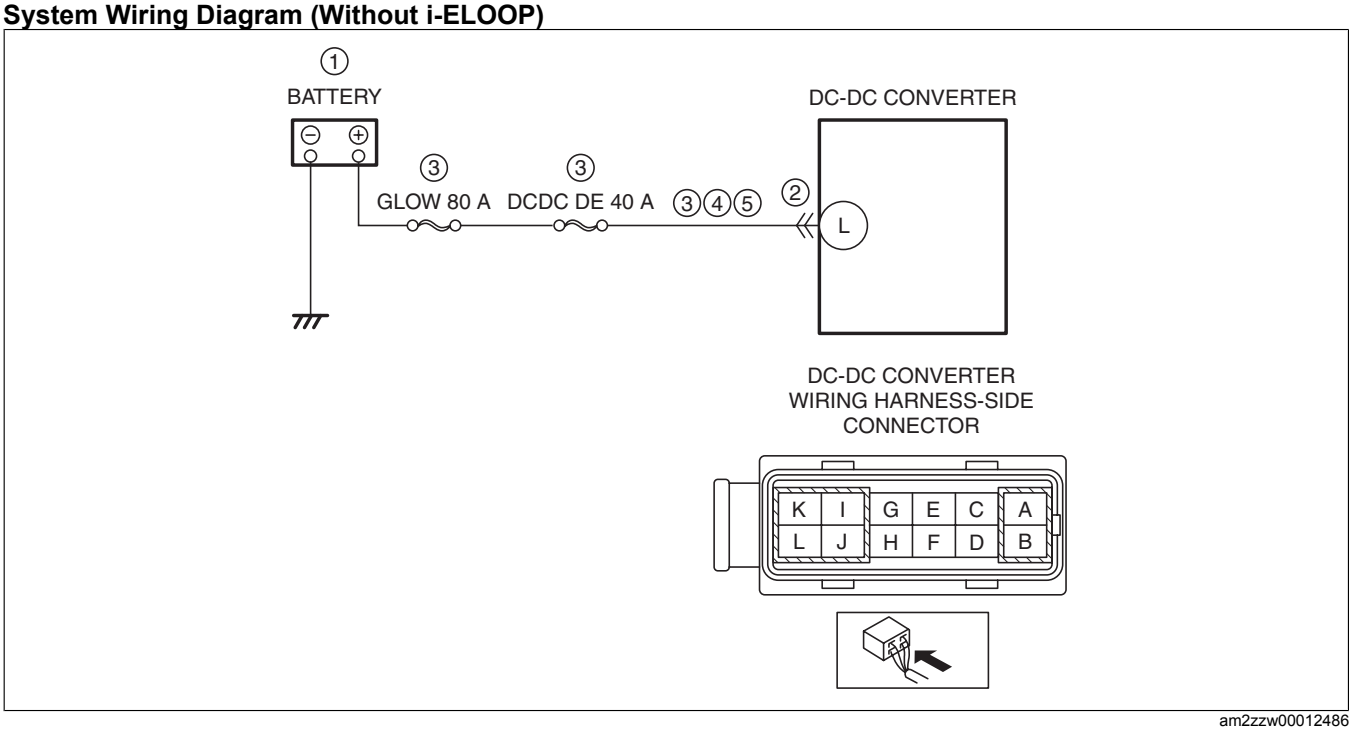
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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs (Without i-ELOOP)

DESCRIPTION	**Power supply system circuit low input**	
DETECTION CONDITION	Determination conditions	• Any one of the following conditions is met: — Battery voltage is **7.25 V or less** when engine restarts from i-stop. — DC-DC converter terminal voltage is **6 V or less** when engine restarts from i-stop. — PCM terminal voltage is **6.2 V or less** when engine starts. — Battery internal resistance is specified value or more. — Battery voltage is **11 V or less** when engine is switched off.
	Preconditions	• Not applicable
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• Current sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • The engine cannot be started or the engine may stall due to battery voltage decrease.	
POSSIBLE CAUSE	• Battery malfunction • Connector or terminal malfunction of the following parts: — Battery — Current sensor — PCM — DC-DC converter • Malfunction in the following fuses: — GLOW 80 A fuse — DCDC DE 40 A fuse • Short to ground in wiring harness between the following terminals: — Battery positive terminal—DC-DC converter terminal L • Open circuit in wiring harness between the following terminals: — Battery positive terminal—DC-DC converter terminal L • Current sensor malfunction • BCM malfunction • DC-DC converter malfunction • PCM malfunction	

System Wiring Diagram (Without i-ELOOP)



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Function Explanation (DTC Detection Outline) (Without i-ELOOP)

- The PCM assures engine startability and power during driving based on control of the battery related parts. The PCM detects the battery voltage during engine starting and the battery internal resistance based on the current sensor signal, and if the battery voltage during engine starting is low or the battery internal resistance is high, it determines that there is a malfunction in the battery related parts and stores a DTC.

Repeatability Verification Procedure (Without i-ELOOP)

1. Start the engine.
2. Perform battery condition initial setting (i-stop setting). (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).)

PID Item/Simulation Item Used In Diagnosis (Without i-ELOOP)

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
BATT_CUR	Battery charge/discharge current	A	• Displays the battery charge/discharge current value.
VPWR	Battery positive voltage	V	• Switch the ignition ON (engine off): Approx. 12.78 V • Idle: Approx. 13.78 V

Function Inspection Using M-MDS (Without i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF POWER SUPPLY IS AFFECTED BY DTC RELATED TO DC-DC CONVERTER** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are DTCs related DC-DC converter recorded?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
3	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CURRENT SENSOR** • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P058A:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P058A:00 [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
4	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CAN OR LIN COMMUNICATION** • Perform the PCM and BCM DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [BCM].) • Are DTCs related CAN or LIN communication recorded?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].) (See DTC TABLE [BCM].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (Without i-ELOOP)
Intention of troubleshooting procedure

- Step 1
  - — Perform a unit inspection of the battery.
- Step 2—5
  - — Perform the inspection for each power supply line.
- Step 6—7
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: DETERMINE INTEGRITY OF BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Replace the battery, then go to Step 6. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Disconnect the connector of the following parts. — Battery — Current sensor — PCM — DC-DC converter • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.

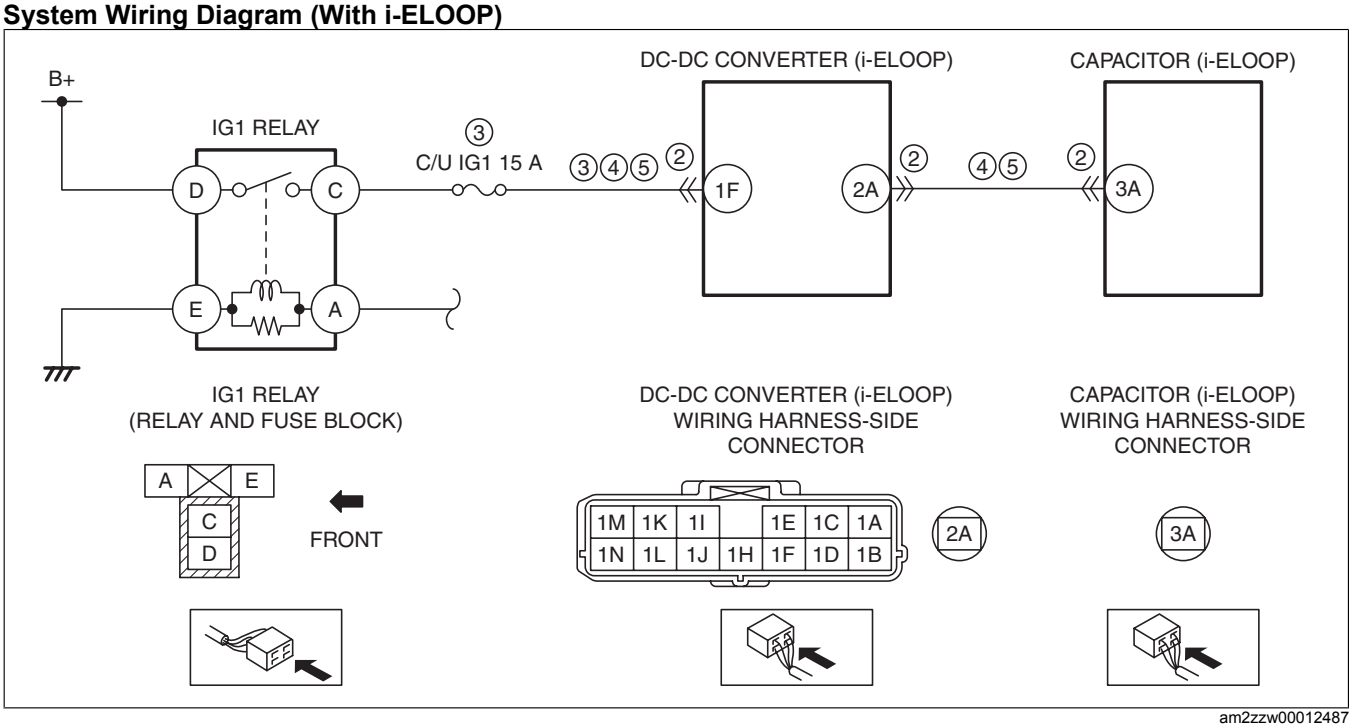
STEP	INSPECTION	RESULTS	ACTION
3	**PURPOSE: INSPECT FUSE** • Remove the following fuses: — GLOW 80 A fuse — DCDC DE 40 A fuse • Inspect the following fuses: — GLOW 80 A fuse — DCDC DE 40 A fuse • Are all the fuses normal?	Yes	Install all the removed fuses, then go to the next step.
		No	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Battery positive terminal—DC-DC converter terminal L **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the malfunctioning fuse. If the fuse is damaged: • Replace the malfunctioning fuse. Go to Step 6.
4	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, current sensor, PCM and DC-DC converter connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — DC-DC converter terminal L • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Battery positive terminal—DC-DC converter terminal L **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 6.
		No	Go to the next step.
5	**PURPOSE: VERIFY IF OPEN CIRCUIT IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, current sensor, PCM and DC-DC converter connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Battery positive terminal—DC-DC converter terminal L • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Battery positive terminal—DC-DC converter terminal L **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
6	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure (Without i-ELOOP).) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.

Details On DTCs (With i-ELOOP)

DESCRIPTION	Power supply system circuit low input	
DETECTION CONDITION	Determination conditions	• Any one of the following conditions is met: — Battery voltage is **7.25 V or less** when engine restarts from i-stop. — PCM terminal voltage is **6.2 V or less** when engine starts. — Battery internal resistance is specified value or more. — Battery voltage is **11 V or less** after **30 s** since system began switch to relay mode
	Preconditions	• Not applicable.
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• Current sensor
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • The engine cannot be started or the engine may stall due to battery voltage decrease.	
POSSIBLE CAUSE	• Battery malfunction • Connector or terminal malfunction of the following parts: — Battery — Current sensor — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Malfunction in the following fuses: — C/U IG1 15 A fuse • Short to ground in wiring harness between the following terminals: — IG1 relay terminal C—DC-DC converter (i-ELOOP) terminal 1F — DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A • Open circuit in wiring harness between the following terminals: — IG1 relay terminal C—DC-DC converter (i-ELOOP) terminal 1F — DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A • Current sensor malfunction • BCM malfunction • DC-DC converter (i-ELOOP) malfunction • PCM malfunction	

System Wiring Diagram (With i-ELOOP)



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Function Explanation (DTC Detection Outline) (With i-ELOOP)

- The PCM assures engine startability and power during driving based on control of the battery related parts.
- The PCM detects the battery voltage during engine starting and the battery internal resistance based on the current sensor signal, and if the battery voltage during engine starting is low or the battery internal resistance is high, it determines that there is a malfunction in the battery related parts and stores a DTC.

Repeatability Verification Procedure (With i-ELOOP)

1. Start the engine.
2. After leaving for **30 s** with the engine idling and no electrical load, leave for **60 s** at the maximum electrical load (**50 A or more**).
3. Verify the i-stop control operation by performing the idle speed control learning and determining the success or failure of the idle speed control learning.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
BATT_CUR	Battery charge/discharge current	A	• Displays the battery charge/discharge current value.
VPWR	Battery positive voltage	V	• Switch the ignition ON (engine off): Approx. 12.78 V • Idle: Approx. 13.78 V

Function Inspection Using M-MDS (With i-ELOOP)

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF POWER SUPPLY IS AFFECTED BY DTC RELATED TO DC-DC CONVERTER (i-ELOOP)** • Switch the ignition off, then ON (engine off). • Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs present?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
3	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CURRENT SENSOR** • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P058A:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P058A:00 [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
4	**PURPOSE: VERIFY IF BATTERY VOLTAGE IS FALSELY RECOGNIZED BY DTC RELATED CAN OR LIN COMMUNICATION** • Perform the PCM and BCM DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [BCM].) • Are DTCs related CAN or LIN communication recorded?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].) (See DTC TABLE [BCM].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure (With i-ELOOP)
Intention of troubleshooting procedure

- Step 1
  - — Perform a unit inspection of the battery.
- Step 2—5
  - — Perform the inspection for each power supply line.
- Step 6—7
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: DETERMINE INTEGRITY OF BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Replace the battery, then go to Step 6. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Remove the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Disconnect the connector of the following parts. — Battery — Current sensor — PCM — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
3	**PURPOSE: INSPECT FUSE** • Remove the C/U IG1 15 A fuse. • Inspect the C/U IG1 15 A fuse. • Are all the fuses normal?	Yes	Reinstall the C/U IG1 15 A fuse, then go to the next step.
		No	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay terminal C and DC-DC converter (i-ELOOP) terminal 1F. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the fuse. If the fuse is damaged: • Replace the fuse. Go to Step 6.
4	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the battery, current sensor, PCM, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected. • Remove the IG1 relay. (See RELAY LOCATION.) • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — IG1 relay terminal C — DC-DC converter (i-ELOOP) terminal 2A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • IG1 relay terminal C—DC-DC converter (i-ELOOP) terminal 1F • DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 6.
		No	Go to the next step.
5	**PURPOSE: VERIFY IF OPEN CIRCUIT IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • IG1 relay is removed. • Verify that the battery, current sensor, PCM, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — IG1 relay terminal C—DC-DC converter (i-ELOOP) terminal 1F — DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • IG1 relay terminal C—DC-DC converter (i-ELOOP) terminal 1F • DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
6	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure (With i-ELOOP).) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.